HARIS P P

Senior Software Developer — Qt/C++ & Python | Desktop, Real-Time Video & Computer Vision Systems

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PROFESSIONAL SUMMARY

Senior Software Developer with 14+ years building desktop, embedded, and computer-vision applications across Windows, Linux, macOS, Android, and iOS. Deep expertise in Qt/C++ and Python for high-performance cross-platform products spanning real-time video streaming (FFmpeg, RTSP, WebRTC, HLS), database-driven desktop applications, and AI/computer-vision systems, including work on medical imaging software. Top-Rated on Upwork with 100% Job Success, owning the full lifecycle from architecture to deployment.

CORE SKILLS

Languages: C/C++, Python,c#, Java, JavaScript, PHP, HTML, Shell Script, Dart

Frameworks & Libraries: Qt 5/Qt 6/QML, PyQt, OpenCV, TensorFlow, Keras, Flutter, WebRTC, FFmpeg, HLS, OpenSSL

Databases: MySQL, SQLite (transaction batching & performance tuning)

Platforms: Windows, Linux (Debian, Ubuntu), macOS, Android, iOS, Raspberry Pi, BeagleBoard

Domains: Computer Vision, Real-Time Video Streaming, Medical Imaging, IoT, Desktop Applications, Data Security/Encryption

PROFESSIONAL EXPERIENCE

Full-Time Freelancer — Self-Employed (Remote) *2023 – Present*

Top-Rated on Upwork · 100% Job Success · 4.8★ · 23 jobs · 454+ hours

- Deliver cross-platform desktop, embedded, and computer-vision solutions for international clients across the US, UK, and Europe.
- **AlgoMedica — PixelShine (AI-Powered CT Scan Noise Reduction Software):** Migrated AlgoMedica's Qt-based medical imaging software from Debian 9 to Debian 12, including Qt code migration, dependency/package upgrades, dev environment setup, and documentation updates for a deep-learning CT scan enhancement product used in clinical radiology.
- **Trading Application — Desktop & Algorithmic Trading Client:** Built Windows desktop trading client in Qt/C++, communicating with a local database and a live exchange socket for real-time market data. Implemented real-time exchange data feeds, new algorithmic strategies for portfolio management, and backend communication layers. Also maintain and extend the client's existing C# trading application with new feature implementations.
- **Proconnecta — Desktop & Audio Streaming Applications (Qt/C++):** Engaged as developer on two client products.

- **Streambox** *(Music Streaming Desktop App)* — Migrated the codebase from Qt 5 to Qt 6 and extended the app from Windows-only to a native macOS build; integrated OpenSSL and other core dependency libraries, added AES encryption for user data, delivered a more professional UI overhaul, and fixed key bugs to improve app stability and speed, including a major login-freeze issue on large libraries.
- **MixWorx** *(DJ & Audio Broadcasting Platform)* — Extended the open-source Mixxx codebase with a custom encrypted music library integration and third-party API integrations for real-time audio mixing and broadcasting.

Senior Software Developer — Vappica *Jun 2014 – Dec 2022*

- Spearheaded development of a network video recorder (NVR) platform in Qt/C++ on Ubuntu, ingesting and recording IP camera streams via FFmpeg and RTSP, with OpenCV-based AI analytics for face detection/recognition, person and vehicle detection, and license plate recognition (LPR).
- Designed and built cross-platform client applications — Windows (Qt/QML), Android, iOS, and Web (HLS) — for live playback, recorded footage search, and analytics review (face recognition, LPR, and detection event browsing).
- Led a WebRTC-based classroom platform with audio/video/text chat, assignments, exams, and student management.
- Built Raspberry Pi camera recorders with detection & alert systems, a WebRTC intercom (Android/iOS clients), and a beacon configuration app for advertisement systems.
- Developed web tooling for access control and remote door operations using Apache, MySQL, and PHP on Ubuntu.

Design Engineer — NiLogic Pvt Ltd *Jun 2011 – May 2014*

- Developed computer-vision applications on BeagleBoard for fire/cloud detection during explosions and dimension measurement.
- Cross-compiled the Linux kernel for BeagleBoard platforms.
- Built an Android application to control Raspberry Pi GPIO functions.

SELECTED PROJECTS

- **Network Camera Recorder with Analytics —** *(C++/Qt · OpenCV · HLS · object/motion detection)*
- **Replay Video Capture — Screen Recording Application (Windows & macOS) —** Desktop screen & video capture application handling high-quality recording on Windows and macOS *(C++/Qt · Windows · macOS)*
- **Teleteens — Parental Control App (Windows) —** *(C++/Qt · monitoring logic & UI)*
- **Embedded Computer Vision (BeagleBoard/Raspberry Pi) —** *(OpenCV · Python/C++ · Embedded Linux)*

EDUCATION

M.Sc. Electronics — Cochin University of Science and Technology

B.Sc. Electronics — Kannur University